

EpiSave EMU: A multi-centre, event-driven clinical study to assess the performance of the EpiSave generalized convulsive seizure detection system in people with epilepsy in Epilepsy Monitoring Units (EMUs)

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Research legislation:	Ordinance on human research with the exception of Clinical trials (HRO) [1].
Type of Research Project:	Research project involving human subjects
Risk Categorisation:	Risk category A acc. to ordinance HRO Art. 7 - observational study with non-invasive device. The alarm will be available only for the medical team and the participant will be blind to whatever the algorithm detects.
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PROTOCOL SIGNATURE FORM

Study Title **EpiSave EMU** - A multi-centre, event-driven clinical study to assess the performance of the EpiSave generalized convulsive seizure detection system in people with epilepsy in Epilepsy Monitoring Units (EMUs)

The project leader at the main coordinating centre and the project leader at the local centre/site have approved the protocol version 1.0 (dated 05.12.2025) and confirm hereby to conduct the project according to the protocol, the Swiss legal requirements [1,2], the current version of the World Medical Association Declaration of Helsinki [3] and the principles and procedures for integrity in scientific research involving human beings.

The project leader at main centre has received the ICF and consider it appropriate for use.

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GLOSSARY OF ABBREVIATIONS

<i>3D-acc</i>	<i>3D-accelerometer</i>
<i>ASM</i>	<i>Anti-Seizure Medications</i>
<i>BASEC</i>	<i>Business Administration System for Ethical Committees</i>
<i>CE</i>	<i>European Conformity</i>
<i>CRF</i>	<i>Case Report Form</i>
<i>EC</i>	<i>Ethics Committee</i>
<i>eCRF</i>	<i>electronic Case Report Form</i>
<i>EEG</i>	<i>Electroencephalogram</i>
<i>EMUs</i>	<i>Epilepsy Monitoring Units</i>
<i>FAR</i>	<i>False Alarm Rate</i>
<i>FDA</i>	<i>Food and Drug Administration</i>
<i>FBTCS</i>	<i>Focal to Bilateral Tonic-Clonic Seizures</i>
<i>GCS</i>	<i>Generalized Convulsive Seizures</i>
<i>GDPR</i>	<i>General Data Protection Regulation</i>
<i>GTCS</i>	<i>Generalized Tonic-Clonic Seizures</i>
<i>HRA</i>	<i>Human Research Act</i>
<i>HRO</i>	<i>Ordinance on Human Research</i>
<i>ICF</i>	<i>Informed Consent Form</i>
<i>iOS</i>	<i>iPhone Operating System</i>
<i>PPG</i>	<i>Photoplethysmography</i>
<i>PPI</i>	<i>Patient and Public Involvement</i>
<i>PWE</i>	<i>People with Epilepsy</i>
<i>SUDEP</i>	<i>Sudden Unexpected Death in Epilepsy</i>
<i>TWP5</i>	<i>TicWatch Pro 5</i>
<i>VEEG</i>	<i>Video Electroencephalogram</i>

1 BACKGROUND AND PROJECT RATIONALE

Epilepsy is a neurological condition affecting approximately 60 million people worldwide. Among those, 25-30% experience generalized convulsive seizures (GCS), which include focal to bilateral tonic-clonic (FBTCS) and generalized tonic-clonic seizures (GTCS) [4-6].

GCS represent one of the most severe forms of epileptic seizures and are associated with increased risks of physical injury, sudden unexpected death in epilepsy (SUDEP), and significant impact on quality of life [7-9].

Several studies have demonstrated that prompt intervention following GCS detection can reduce seizure duration, prevent secondary complications, and potentially decrease SUDEP risk by up to 40% [10]. Medical devices such as applications that detect GCS and alert caregivers with the patient's location have been developed to enable timely intervention. However, current medically certified seizure detection solutions present significant limitations including high false-alarm rates (FARs), with more than one false alarm per week, substantial costs (\$250-1,500 USD for device purchase plus monthly subscription fees), limited operational windows (night-only for some systems), and stigmatization concerns related to dedicated medical devices.

Currently Available Medical Devices:

1. **EpiMonitor (Empatica Inc.):** FDA (Food and Drug Administration)-cleared, multimodal GTCS detection device utilizing electrodermal activity, accelerometry (3D-acc), and photoplethysmography (PPG) sensors. Clinical validation studies demonstrate a detection accuracy of 98% compared to hospital-based electroencephalogram (EEG) measurements. Approved for adults and children ages 6 years and older [11, 12].
2. **NightWatch (LivAssured B.V.):** A nocturnal seizure detection system utilizing heart rate variability and 3D-acc for GTCS detection. Clinical validation demonstrates a sensitivity of 86% with a FAR of 0.7 per night. Its use is limited to nocturnal monitoring only [13-15].
3. **Epi-Care Mobile (Danish Care Technology):** A mobile seizure detection system utilizing 3D-acc for convulsive seizure detection. Clinical validation studies show a variable sensitivity ranging from 65% to 90%, depending on seizure type and patient population [16, 17].
4. **EpiWatch:** FDA-cleared iOS (iPhone Operating System) app running on Apple watch detects GTCS using 3D-acc only. Studies showed a sensitivity of 100% with a FAR of 0.05 per day [19].

These limitations collectively result in low adoption rates among people with epilepsy (PWE), with only a small minority having access to or choosing to use existing certified seizure detection devices due to cost barriers, usability concerns, and lifestyle restrictions imposed by current solutions.

EpiSave addresses these limitations through an integrated system composed of three elements: the EpiSave algorithm, the EpiSave smartphone app and the EpiSave smartwatch app. Together, these components form the EpiSave system, which is designed to detect GCS with high sensitivity (preliminary data indicate a 96% detection rate), while maintaining a low false alarm rate (<1 per week). The system operates on commercially available low-cost smartwatches that can serve other functions and help reduce stigma, provides immediate alerts to caregivers including patient location, and offers the application free of charge to PWE in low- and middle-income countries.

The development of seizure detecting devices and applications follows a standardized framework. EpiSave has successfully completed phase 2 testing [18], which involved algorithm development and offline validation on 3D-acc data previously collected in EMU where video-EEG (VEEG) serves as a gold-standard. EpiSave algorithm was first trained on data from 37 participants who suffered 54 GCS. It was then tested on an independent dataset from 347 participants, including 33 with 49 GCS, showing a 96% sensitivity (95% CI 90%-100%) and a FAR <1 per week [19].

The present study represents a phase 3 validation study in the EMU setting, with VEEG serving as the gold standard reference for seizure detection.

This research project is classified as Category A under HRO Art. 7, as it is an observational study using a non-invasive device. The alarm will be accessible exclusively to the medical team and will not interfere with standard clinical procedures. Participants will remain unaware of any algorithmic detections. The intervention—wearing a smartwatch—is non-invasive, and continuous monitoring does not alter standard care or impose any additional burden beyond routine EMU monitoring. Consequently, participants will not experience any constraints or discomfort related to this study.

2 PROJECT OBJECTIVES AND DESIGN

2.1 Hypothesis and primary objective

The EpiSave seizure detection algorithm is hypothesized to achieve a sensitivity of at least 80% (lower 95% confidence interval end) for detecting GCS (including GTCS and FBTCS) when validated against the VEEG gold standard in EMU settings.

Primary Objective: To assess the EpiSave algorithm sensitivity for detecting GCS (including GTCS and FBTCS) in EMU settings validated against VEEG gold standard.

Secondary Objectives:

- 1) To evaluate the EpiSave algorithm FAR in EMU settings and analyse factors influencing false alarms, including participants activities and contextual triggers.
- 2) To evaluate the delay between GCS onset and its detection.
- 3) To test the performance of the EpiSave algorithm in characterizing GCS such as the precision of the duration of GCS and post-ictal immobility.
- 4) To assess the reliability and usability of the EpiSave system in a controlled environment a custom satisfaction survey will be completed by the study nurse, not by the participant.
- 5) To document any technical issues or limitations encountered with the EpiSave system (any adverse event).
- 6) To assess smartwatch wearability and comfort during EMU stay, including tolerance (skin irritation, discomfort).
- 7) To evaluate the performance of the EpiSave algorithm when applied to data collected using a smartphone worn on the upper arm during nighttime.

2.2 Primary and secondary endpoints

	Definition	Measurement	Analysis
Primary Endpoint			
EpiSave algorithm sensitivity for detecting GCS validated against VEEG recording in EMU	Proportion of GCS detected by the EpiSave algorithm among all GCS confirmed by VEEG	$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \times 100\%$	95% confidence intervals
Secondary endpoint			
False alarm rate	Proportion of false positive detections per participant per day	Count of false alarms divided by monitoring days and reporting the	95% confidence intervals



		reason of the false alarm such as teeth brushing etc.	
System Uptime	Proportion of monitoring time with active EpiSave system	Continuous monitoring logs	Mean uptime percentage
Technical Performance	Device performance metrics: battery life, connection reliability statistics	Automated system logs	Descriptive statistics
User experience	Staff satisfaction (=caregivers) with the EpiSave system	Custom satisfaction survey	Mean scores with standard deviation
GCS Characterization	Accuracy of GCS characterization	Comparison with VEEG analysis	Descriptive statistics
Smartwatch Wearability and Comfort	Tolerance and comfort during EMU stay, including skin irritation, discomfort	Participant-reported observations and short questionnaire at the end of EMU stay	Descriptive statistics
EpiSave algorithm sensitivity for detecting GCS with data collected by the smartphone at the upper arm against VEEG recording in EMU	Proportion of GCS detected by the EpiSave algorithm with the data collected by the device at the arm among all GCS confirmed by VEEG	$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \times 100\%$	95% confidence intervals

The study will collect a comprehensive set of variables from participants, including demographic characteristics, epilepsy history, seizure types and frequencies, suspected seizure onset side and localization, etiologies, comorbidities, current and past antiepileptic drugs, seizure-related symptoms detectable by wearable devices (e.g., motor signs, tachycardia, falls), and study-specific data such as EMU stay information and wearable device usage. These variables represent factors that may influence the primary and secondary endpoints and should therefore be considered in the data analysis.

2.3 Independent variables

Primary Independent Variable: EpiSave algorithm

The primary independent variable is the EpiSave algorithm, a proprietary machine learning algorithm developed for the detection of GCS. It operates based on continuous analysis of 3D-acc data and is designed to function throughout the entire EMU stay.

Secondary Independent Variable: The android smartwatch

The secondary independent variable is a commercially available Android smartwatch running Wear OS 4.0, equipped with a 3D-acc operating at a 27 Hz sampling rate. Data collected by the device is stored locally and securely uploaded to a dedicated server. The seizure alert system is deactivated for participants, who remain unaware of any detections. Alerts are transmitted exclusively to EMU staff, who are responsible for verifying GCS occurrences, documenting false positives (e.g., no GCS despite detection), and reporting false negatives (e.g., GCS not detected by the system).

The third independent variable: The smartphone worn on the upper arm

The third independent variable is a smartphone worn on the upper arm. This motion detector operates using 3D-acc, gyroscope, barometer and etc. The data are stored locally and securely transferred to the NeuroTech CHUV server.

2.4 Project design

The EpiSave EMU study, coordinated by the CHUV, is a multicentre, event-driven, single-arm, open-label, prospective observational validation study conducted in Swiss and European EMUs.

This observational validation study will be conducted across 17 epilepsy centres in Switzerland (CHUV, HUG, InselSpital, USB, UKBB, KISPI, SEC, UZB) and Europe (Dianalund and Copenhagen – Denmark; Lyon – France; Bonn, Frankfurt and Berlin – Germany; Oslo – Norway; Salzburg and Vienna – Austria), all equipped with EMUs. The observational design is appropriate for validating the performance of the EpiSave algorithm against the gold standard of VEEG, without altering standard patient care. The prospective nature of the study allows for real-time data collection from the beginning to the end of each participant's VEEG recording. The multicentre approach ensures generalizability across diverse clinical settings and patient populations.

Participants will be patients with pharmacoresistant epilepsy admitted to the EMU for VEEG monitoring as part of their routine diagnostic evaluation or follow-up. Each participant will wear a wear OS Android smartwatch such as the TicWatch Pro 5 (TWP5) equipped with the EpiSave smartwatch app and may optionally wear a smartphone on the upper arm (specifically on the bicep) on the same side as the smartwatch for a period ranging from 3 days to 2 weeks (average: 5 days), corresponding to the duration of their VEEG recording. This duration depends on the occurrence of seizures, which determines when the clinical assessment can be concluded. The study participation period will not exceed the duration of the clinically indicated VEEG monitoring.

The study includes continuous monitoring throughout the EMU stay, combining real-time GCS detection and post-ictal biosignals recording via the smartwatch and may optionally include continuous recording of the patient's movements with a smartphone on the upper arm. The detection process remains silent for participants, as no alerts are delivered to them. In parallel, patients undergo standard VEEG monitoring as part of their routine clinical care, entirely independent of the research intervention.

EpiSave EMU is an event-driven study, meaning its completion depends on the occurrence of a predefined number of clinical events. Specifically, the study will conclude once 60 GCS events have been recorded across 40 different PWE. The event-driven design enables efficient recruitment until enough GCS are captured. The total duration of the study, encompassing both recruitment and data analysis phases, will not exceed 3 years.

2.5 Patient and public involvement (PPI)

No patients or members of the public were involved in the design of the study protocol, the choice of outcomes, or the planning of study-related procedures. The study was designed exclusively by the research team based on prior evidence and clinical experience. This decision was made because the current study is a technical validation in EMU, where procedures are standardized and determined by clinical requirements, leaving limited scope for patient-driven modifications. Future phases of the EpiSave project will include patient and caregiver input to optimize usability and acceptability of the device in real-world settings.

3 PROJECT POPULATION AND STUDY PROCEDURES

3.1 Project population, inclusion and exclusion criteria

The study will enrol both adults and children aged 6 years and older with drug-resistant epilepsy, defined as persistent seizures despite trials of at least two appropriate anti-seizure medications (ASM). Participants must be at risk of experiencing GCS during their stay in the EMU. The study includes vulnerable individuals such as who may lack the capacity to provide independent consent and the children. This is an event-driven study, aiming to capture a minimum of 60 GCS events across at least 40 PWE. Based on data from the SEVERITY study (2019-00437 – 398 participants, 84 individuals with 121 GCS), we estimate that approximately 235 participants will be recruited to achieve this target, with 40 participants expected to experience at least one recorded GCS. Recruitment will be conducted to ensure balanced representation across sexes and genders, recognizing that epilepsy affects both males and females and that seizure characteristics may vary accordingly. Recruitment will be competitive between sites, meaning that each centre may enrol participants until the overall target sample size is reached, without predefined quotas per site. This approach allows for flexibility and accelerates enrolment, particularly in centres with higher patient throughput.

Patients will be recruited according to the following eligibility criteria:

Inclusion Criteria	Exclusion Criteria
• Age ≥ 6 years old • Confirmed diagnosis of epilepsy (documented by a neurologist) • Drug-resistant epilepsy (persisting seizures despite two adequate ASM trials) • Signed informed consent form from patient or legal representative • PWE at risk of suffering a GCS during VEEG monitoring, either due to: - Documented history of GCS (from medical records or patient/caregiver reports) - Planned tapering of ASM during the monitoring • Minimum planned EMU stay of 3 days	• Age < 6 years old • Uncertain epilepsy diagnosis • Inadequate ASM trials • No GCS risk during VEEG monitoring • Lack of signed informed consent form • Inability to wear smartwatch device due to physical limitations • Known allergy to smartwatch materials (silicone, metal components)

3.2 Recruitment, screening and informed consent procedure

Participant recruitment will be conducted continuously throughout the study. Only patients admitted to the EMU for a VEEG monitoring will be considered for inclusion. Recruitment will take place at participating clinical centres across Switzerland and Europe.

Screening will be conducted during the EMU admission process. No additional procedures beyond standard EMU care will be required. Eligibility will be assessed based on predefined inclusion and exclusion criteria.

Eligible candidates will be identified during their EMU hospitalization. During an explanatory meeting, the project leader explains to each participant the nature of the research project, its purpose, the procedures involved, the expected duration, the potential risks and any discomfort it may entail. Each participant is informed that the participation in the research project is voluntary and that he/she may withdraw from the research project at any time and that withdrawal of consent will not affect his/her subsequent medical assistance and treatment. The participants are informed that he/she can ask any question. Participants are given sufficient time to review the information and make an informed decision regarding participation. To maximize the likelihood of capturing potential GCS during the EMU stay, monitoring with the smartwatch and may optionally include the motion sensor from the smartphone worn on the upper arm, should ideally begin as soon as possible, preferably within 24 hours of admission, recognizing that not all patients will remain hospitalized for the entire week.

All participants are given an information document and a consent form describing the research project. The formal consent of a participant, using the approved consent form, is obtained before the participant is enrolled in the research project. The participant should read, understand, and voluntarily agree before signing and dating the informed consent form (ICF), and is given a copy of the signed document. The consent form is signed and dated by the participant and the project leader (or her/his designee). The signed consent form it is retained as part of the investigation records.

For paediatric participants, age-appropriate information will be provided. Children aged 6 to 13 years will receive both verbal and written information adapted to their developmental level. Adolescents aged 14 and older will receive the same information as their parents or legal representatives and will be asked to sign the consent form themselves.

No monetary compensation will be provided to participants.

3.3 Study procedures

The study is event-driven and anticipated to last up to 36 months, including recruitment period of 30 months and data analysis period of 6 months.

The site's physician-investigator and co-investigator are responsible for screening patients upon admission to the EMU by verifying inclusion and exclusion criteria, to identify potential candidates for study participation. The duration of each participant's involvement in the study corresponds to that of standard VEEG monitoring—typically around 5 days, with a minimum of 3 days and a maximum of 2 weeks. No additional procedures beyond routine clinical care are performed as part of the study.

During the first visit (V1), upon admission to the EMU, the principal investigator or co-investigator will explain the study's objectives and procedures to the patient. The patient will be given as much time as needed to review the information and sign the ICF. Once the consent is obtained, the investigator will collect demographic and clinical data through a brief interview and review of the patient's electronic medical records. Prior to the enrolment of the first participant, EMU staff will receive training on the Wear OS Android smartwatch device, the EpiSave system and the smartphone used to collect upper arm motion data.

During the monitoring period in the EMU, the participant will wear a Wear OS Android smartwatch on their non-dominant wrist and may optionally wear a smartphone on the same arm (specifically at the level of the biceps), alongside standard VEEG recording. This setup enables continuous monitoring via the EpiSave smartwatch app, allowing direct comparison between smartwatch data and the gold-standard VEEG. The EpiSave system detects and records GCS in real time; however, no alerts will be sent to the participant, who will remain unaware of any detections. Alerts will be transmitted exclusively to the nursing staff via the paired smartphone running the EpiSave smartphone app. In addition to GCS detection, the smartwatch will record biosignals—such as photoplethysmography and gyroscope data—for 15 minutes following each

detected GCS. Wearing the smartphone allows continuous measurement of movements by using 3D-acc, a gyroscope, barometer etc at upper arm level. The data recorded by the smartphone will be transferred to the NeuroTech CHUV server for analysis by the EpiSave algorithm, in order to compare performances from smartwatch, data collected on the wrist vs smartphone data collected on the upper arm. Daily checks will be performed to ensure proper device functioning, including battery status and connectivity. Data collected by the smartwatch will be automatically uploaded to the Exoscale server in Switzerland when the device is charging and connected to Wi-Fi or Bluetooth via the provided smartphone and an Internet connection. Data collected by the smartphone will be retrieved and uploaded NeuroTech CHUV server in Switzerland.

The EpiSave algorithm was developed, trained, and validated using data from the SEVERITY study (2019-00437), which combines wearable sensor data with VEEG recordings, enabling precise event labelling for model development. The model is an ensemble of sub-models, each trained on a distinct subset of the training data. All sub-models share the same architecture—a convolutional neural network followed by a multi-layer perceptron—and process 30-second segments of motion data. Each sub-model outputs a confidence score between 0 and 1, representing the likelihood of a seizure. The final decision score is the median output from the sub-models, with a median score above 0.5 indicating a detected GCS.

If the EpiSave algorithm detects a GCS that is not confirmed by VEEG review (false positive), or if a GCS is missed by the EpiSave algorithm but identified in the VEEG (false negative), EMU staff must document these events both in the EpiSave smartphone App (EpiSave e-diary) and in the dedicated REDCap eCRF form. For false positives, the staff should report the motor activity that triggered the alert. For false negative, a GCS entry must be manually added to the e-diary. The app records the date and time of each alert, whether a true GCS was confirmed by VEEG, and—if no GCS occurred—details about the patient's activity or context that may have led to the false positive. Corresponding video recordings (true GCS, false positives, and false negatives) will be archived for subsequent analysis.

No other action will be taken regarding the participant's medical care unless the physician considers that the patient's safety is at risk.

At the end of the VEEG recording (Visit 2), the smartwatch and, if worn, the smartphone will be retrieved. The patient will be asked to complete one or two short questionnaire(s) about the wearability and comfort of the smartwatch and, if applicable, the smartphone worn on the arm. The EMU nurse/technician will be asked to complete a short custom EpiSave smartphone app satisfaction survey after patient's discharge.

The wear OS Android smartwatch

The wear OS Android smartwatch used in this study is a commercial CE (European Conformity)-marked device capable of measuring various biosignals. For the purpose of this study, only the 3D-acc data will be collected continuously throughout the monitoring period. The other signals—PPG and gyroscope data—will be recorded exclusively during the 15 minutes following a detected GCS.

The EpiSave smartwatch app is pre-installed on the smartwatch to detect GCS and send alerts to EMU nursing staff via the paired smartphone. The device will be fully configured and ready for use before the first study visit (V1). The participant will begin wearing the smartwatch immediately after it is handed over and ICF signed. At that point, the EpiSave system will be active and ready to record data.

The smartwatch battery allows for approximately >24 hours of data recording and requires less than 2 hours for a full recharge. The nursing staff will check the battery status regularly and recharge the device when needed. To ensure uninterrupted data collection during charging periods, a second pre-configured smartwatch will be provided to the participant. Further details on the smartwatch functionality with EpiSave is described in appendix 2.

Smartphone and Armband Setup

The smartphone used in this study will be secured on the participant's upper arm using an adjustable armband. This configuration enables continuous monitoring of arm movements throughout during each night of the study period. Motion data will be collected via the smartphone's integrated sensors, including the barometer, 3D-acc, and gyroscope.

A dedicated study application will be pre-installed on the smartphone to record sensor data and ensure proper synchronization with the study database. The device will be fully configured and ready for use before the first study visit (V1). The participant will begin wearing the smartphone at the first night in EMU, immediately after it is handed over and the ICF is signed. The smartphone battery allows for approximately >24 hours of continuous data recording and requires less than 2 hours for a full recharge. Nursing staff will recharge the smartphone during the day to ensure it is fully charged and ready for use each night.

3.4 Withdrawal and discontinuation

Participants will be withdrawn from this study for the following reasons:

- Withdrawal of informed consent
- Lack of compliance
- Upon participant request at any time
- Changes in health condition leading to the appearance of an exclusion criterion
- Device intolerance or adverse reactions

In case of withdrawal from the study, coded data of withdrawn participant will remain for analysis. The data collected until withdrawal (eCRF, device and app) will be used in a coded manner to avoid compromising the scientific validity of the study. Participant's care and medical follow-up will not be affected in case of withdrawal.

4 STATISTICS AND METHODOLOGY

4.1. Statistical analysis plan

Sample size calculation was conducted in accordance with established guidelines for the clinical validation of seizure detection devices [18]. Based on our experience from the SEVERITY study (398 participants, 84 with 121 GTCS), we estimate that 60 GCS would require recruiting 235 participants, with 40 expected to have at least one recorded GCS.

Power Calculation

The sample size calculation was based on the primary endpoint, which is the sensitivity for GCS detection. The target sensitivity was set at $\geq 80\%$, while the expected sensitivity, derived from preliminary data, is 96%. Assuming a two-sided significance level of $\alpha = 0.05$ and a statistical power of 80%, the analysis indicated that at least 60 GCS events are required to achieve adequate precision for the primary endpoint.

Statistical Methods

Statistical methods will include a primary analysis of sensitivity with 95% confidence intervals calculated using Wilson's method, and secondary analyses covering false alarm rate per participant per day, system reliability metrics, EMU staff experience measures, GCS characterization accuracy, smartwatch tolerance and comfort during the EMU stay and EpiSave algorithm sensitivity with data from data collected from the smartphone, all summarized using descriptive statistics; analyses will be performed using Python or R (version 4.2.0 or later).

Level of Significance

The level of significance for the primary endpoint will be two-sided at $\alpha = 0.05$, and no adjustment for multiple testing will be applied since all secondary endpoints are considered exploratory.

4.2. Handling of missing data

Missing data may occur due to incomplete EMU stays, technical failures during monitoring, or missing questionnaire responses. To address this, missing questionnaire data will be handled using multiple imputation with chained equations. Sensitivity analyses will be performed to evaluate the impact of missing data on primary outcomes. Both per-protocol analyses, including only participants with complete data, and intention-to-treat analyses, including all enrolled participants, will be conducted.

To minimize missing data, several preventive measures will be implemented: robust device testing and backup systems, standardized data collection procedures, regular quality control checks, and participant education and support throughout the study.

5 REGULATORY ASPECTS AND SAFETY

5.1 Local regulations / Declaration of Helsinki

This research project will be conducted in accordance with the protocol, the Declaration of Helsinki [3], the Human Research Act (HRA) and the Human Research Ordinance (HRO) [1] as well as other locally relevant regulations. The project leader acknowledges his responsibilities as both the project leader and the Sponsor.

5.2 Notification of safety and protective measures (HRA Art. 15, HRO Art. 20)

If, during the research project, circumstances arise which could jeopardise the safety or health of the participants or lead to a disproportionate relationship between the risks and burdens and the benefits, all the measures required to ensure protection are to be taken without delay.

The project leader is promptly notified (within 24 hours) if immediate safety and protective measures must be taken during the conduct of the research project. The Ethics Committee will be notified via BASEC (Business Administration System for Ethical Committees) of these measures and of the circumstances necessitating them within 7 days.

5.3 Serious events (HRO Art. 21)

If a serious event occurs, the research project will be interrupted and the Ethics Committee (EC) notified on the circumstances via BASEC within 7 days according to HRO Art. 21.

The project leader reports to the EC on the connection between the event and the collection of health-related personal data. At the same time, the project leader submits proposals concerning the next steps to be taken.

Any new relevant information and the outcome to the original Serious Event is reported to the EC via BASEC.

5.4 Procedure for investigations involving radiation sources

This study does not involve radiation sources. No additional procedures are required.

5.5 Amendments

Substantial changes to the project set-up, the protocol and relevant project documents will be submitted to the EC for approval according to HRO Art. 18 before implementation. Exceptions are measures that have to be taken immediately in order to protect the participants.

The following are considered to be substantial changes:

- a. changes affecting the participants' safety and health, or their rights and obligations;
- b. changes to the protocol which concern the objectives of the research project;
- c. a change of research site or conducting the research project at an additional site; or
- d. a change of project leader or Sponsor.

5.6 End of project

Upon project completion or discontinuation, the EC is notified within 90 days. The completion of the research project is defined by the last collection of health-related personal data. All data collected are coded, with the exception of consents, which are treated as confidential. Research data will be stored during at least 10 years after research completion.

5.7 Insurance In the event of study-related damage or injuries, the liability of the CHUV provides compensation.

6 FURTHER ASPECTS

6.1 Overall ethical considerations

This study addresses a significant unmet need in epilepsy care by evaluating an innovative and affordable GCS detection solution, with the potential to greatly improve access to reliable GCS detection for millions of individuals living with epilepsy worldwide.

EpiSave EMU is a prospective observational study involving patients hospitalized in the EMU. Data are collected using one or two non-invasive wearable device(s), without sending any alerts to the participant in the event of a GCS detection. The study does not expose participants to any additional risk, nor does it require extra time or effort on their part. The EpiSave EMU study presents minimal risks while offering potential benefits to the broader epilepsy community. Its observational design ensures that standard clinical care remains unchanged, and the smartwatch intervention and smartwatch is both discreet and minimally invasive.

The multicentre design of the study ensures that the results are generalizable across various EMU settings and patient populations. The study is expected to provide robust evidence to support regulatory approval and future clinical implementation.

All participants, including vulnerable individuals such as children and those with cognitive impairments, receive an information sheet and a consent form that clearly describe the study and provide sufficient details to support an informed decision regarding participation. For minors, consent is obtained from their legal guardians, and assent is sought from the child when appropriate. For participants with impaired decision-making capacity, consent is obtained from a legal representative.

The consent process includes simplified materials and additional support to ensure participant understanding. Consent or assent must be given voluntarily and is documented by the signature and date of the participant, their legal representative or parent, and the principal investigator or designated study staff. A copy of the signed consent form is provided to the participant or their representative, while the original is retained in the study records.

Participants who do not wish to take part in the study are not enrolled. Additional safeguards are implemented for vulnerable populations, including simplified consent procedures, enhanced monitoring, and regular reassessment of consent or assent throughout the study.

Participants retain the right to withdraw from the study at any time and may request information regarding the progress and results of the study.

6.2 Risk-Benefit Assessment

The EpiSave EMU study is observational. To the best of our knowledge, participants are not exposed to any risk throughout the study, as they will be wearing a commercially available CE-marked Android smartwatch and optionally a smartphone. The only potential risk is minor skin irritation from wearing the devices. To mitigate this, regular skin checks will be conducted. This research does not involve any additional examinations and does not interfere with the standard care provided in the EMU.

Concerning privacy risks, the watch will record signals non-invasively and will send recorded data automatically to the Exoscale server when placed on its charger and connected to a Wi-Fi network or Bluetooth to the smartphone and an active Internet connection (further details on the Exoscale server and data handling are provided in section 7.2). To ensure participant data confidentiality, encrypted data transmission, secure storage, and limited access will be implemented. The smartphone is optional for the patient and will record signals non-invasively also. Data are stored locally on the smartphone and will be retrieved by the team and transferred to the secure NeuroTech CHUV server.

No direct individual benefits are anticipated for study participants. However, the results may contribute to the development of improved GCS detection technologies. The validation of the EpiSave system could potentially enhance future access to GCS detection for patients with epilepsy.

The minimal risks are justified by the potential societal benefit of improved GCS detection technology. The study does not modify standard clinical care and provides additional monitoring without additional burden.

6.3 Rationale for the inclusion of vulnerable participants

The EpiSave EMU study includes minors (ages 6–17) and adults with cognitive impairments who may lack the capacity to provide independent consent. These groups are considered vulnerable populations in clinical research. PWE who present with neurobehavioral comorbidities or cognitive dysfunction requiring a legal representative are at increased risk of GCS and SUDEP compared to the general epilepsy population. Due to these impairments, they may be unable to alert caregivers about an impending or past seizure, or to reposition themselves safely following a GCS. This underscores the importance of GCS detection devices for these populations and justifies their inclusion in the study.

Similarly, young children may be at increased risk, as they often lack the reflex to notify caregivers when a seizure occurs and may be unable to reposition themselves afterward. Moreover, equivalent findings cannot be obtained through other means, as these populations exhibit unique seizure characteristics and needs that must be studied directly.

To protect vulnerable participants, the study includes several safeguards: consent is obtained from legal representatives for individuals lacking capacity; assent is sought from minors in addition to legal consent; simplified consent forms are provided for participants with cognitive limitations; monitoring and support are enhanced throughout the study; and ongoing consent and assent are regularly reassessed.

7 QUALITY CONTROL AND DATA PROTECTION

7.1 Quality measures

The project personnel will be trained on all important project related aspects, planned quality visits or independent data review. Standardized training will be provided across all participating centres, including investigator training on protocol procedures and technical staff training on EpiSave system setup and troubleshooting. For quality assurance the EC may visit the research sites. Direct access to the source data and all project related files and documents must be granted on

such occasions. The project leader has appropriate knowledge and skills in the areas of data security and data protection or is able to ensure compliance by calling in appropriate expertise (Art. 4 HRO).

7.2 Data recording and source data

The ICFs, which contain identifiable patient information will be stored in a locked filing cabinet at the local study site where the participant was enrolled. Data storage and access are under the responsibility of the local project leaders and designated study personnel.

Demographic and clinical data will be collected during participant interviews and extracted from the electronic medical record. After coding, these data will be entered into an electronic Case Report Form (eCRF) created in the REDCap database hosted on secure CHUV servers and managed by the CHUV Clinical Research Center (CRC) (see: <https://www.project-redcap.org>). Each participating centre will be provided with a secure access allowing direct entry of data into REDCap. EMU nurses/technicians will also complete a custom EpiSave smartphone app survey to check if everything went well with the EpiSave system and to identify any potential issues or areas for improvement. In addition, participants will complete a questionnaire on smartwatch and smartphone (if worn) comfort during nighttime. REDCap will serve as the official source data for questionnaires and Case Report Forms (CRFs). If paper versions are used as a backup, they will be transcribed into REDCap within 7 days and securely stored in a binder designated for study-related patient data. Queries raised by the coordinating centre (CHUV) will be resolved in REDCap only.

Handling of True Positives, False Positives, and False Negatives

Event Type	Actions in the EpiSave smartphone app	Actions in the REDCap eCRF
GCS confirmed by VEEG (true positives and false negatives)	If false negative: Manually enter GCS in EpiSave smartphone app with date/time and possible reason for non-detection (e.g., smartwatch not worn, technical issue)	- If false negative: Indicate the date and time of the event and possible reason for non-detection (e.g., smartwatch not worn, technical issue) - If true positive: Indicate the date and time of the event
False positives (detected by the EpiSave system but not confirmed by VEEG)	Mark as false positive and specify the activity that may have triggered the alarm	Record the false positive with date and time as well as the activity that may have triggered the alarm
Source of Truth	For all events, REDCap entries will serve as the official source of data. Information entered in the EpiSave smartphone app is considered secondary and may be incomplete due to technical issues.	

The REDCap system ensures secure data entry with controlled access, user rights management, and an audit trail for all modifications, thereby guaranteeing data traceability and integrity throughout the study.

Data from the smartwatch will be collected during the participant's stay in the EMU. These data are automatically transferred to a secure Swiss server (Exoscale) when the smartwatch is placed on its charger and connected to a Wi-Fi network or Bluetooth to the paired smartphone.

For patients who have agreed, motion data from the smartphone will be collected also during the participant's stay during nighttime (sleep) in the EMU. These data are stored locally on the smartphone and will be retrieved by the team and transferred to the secure NeuroTech CHUV server. Smartwatch data will reach the Exoscale server in the form of a unique ID, and will be linked to the subject ID code (see 7.3. for subject ID code generating). Once the upload is done, all temporarily stored data on the device are automatically deleted.

The CHUV study personnel will maintain password-protected excel file with the coding key for the patient study ID on a firewall secured CHUV desktop. The study personnel will regularly check that data upload were transferred towards the secured Exoscale server successfully.

The Exoscale server complies fully with European GDPR regulations. It allows secure streaming of mobile health data without routing through third-party cloud providers, thereby enhancing data protection and giving investigators full control over data flow.

Upon site's principal investigator request, data from the smartwatches acquired within the participants from their centre can be made available. However, data from other sites will not be shared.

Data acquired from the VEEG (true events, false positive and false negative) will be copied from the locally used devices, coded with the participant study ID, encrypted and stored on an appropriate media. Staff from the coordinating centre (CHUV) will physically retrieve the media containing the VEEG files and personally take the data back to the CHUV NeuroTech IT Server.

7.3 Confidentiality and coding

Project data will be handled with utmost discretion and is only accessible to authorized personnel who require the data to fulfil their duties within the scope of the research project. On the CRFs and other project specific documents, participants are only identified by a unique participant number. Coding is done using a method based on the current state of the art that must be based on the current state of the art (Art. 26 HRO).

Health related personal data captured during this project from participants are strictly confidential and disclosure to third parties is prohibited; coding will safeguard participants' confidentiality.

The complete project raw data from the smartwatches and smartphone motion data shall be handled with utmost discretion and will only be accessible to the study personnel at the CHUV. Access to the protocol is reserved to the project partners, and other individuals who are directly implicated.

A password-protected Excel-based log will be prepared at each recruiting centre, called "subject information", which contains the participant names, dates of birth and attributes the subject ID code via the formula, study ID + centre acronym + participant number (e.g. EPE-CHV-001 for CHUV, EPE-HUG-001 for HUG, etc.). This Excel-based log remains confidential at the centre and is not shared with the other centres.

Only the investigators at each site will have access to the identities of the participants enrolled at their respective centres. Access to subject ID codes, both during and after the study, will be restricted to locally involved study personnel.

The code may only be broken if it is necessary to avert an immediate risk to health of the person concerned, to guarantee the rights of the person (e.g. in revoking the consent), or if a legal basis exists for breaking the code.

7.4 Retention and destruction of project data

The project leader retains all the research project data for a period of at least ten years after the completion or early termination of the research project. The study source data generated by the watch will be kept in an archive on the CHUV secure IT servers. Once the study is concluded, the data will immediately be deleted from the Exoscale server (after analysis). Patients will be asked to sign a consent for additional further use of their data.

Study electronic files will be password protected and saved in a secure directory on each site's institutional server.

This study does not involve collection of biological material.

8 FUNDING / PUBLICATION / DECLARATION OF INTEREST

Funding Sources: [TO BE DETERMINED - Funding sources will be disclosed]

Publication Policy: Results will be published regardless of outcome in peer-reviewed journals. Authorship will follow ICMJE guidelines. All investigators will have opportunity to review and comment on manuscripts before submission.

Data Sharing Policy: Coded data may be shared with qualified researchers for secondary analysis, subject to appropriate data sharing agreements and ethics approval.

Declaration of Interest: No conflicts of interest are currently identified. All investigators will promptly update the declaration if any changes occur during the study.

9 REFERENCES

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Appendix 1: Schedule of assessments

Time (days)	Day 0	Day 0 (V1)	Day 0 to EMU discharge	Day 2 to 14 (V2)
Assessment	Information	Screening	Daily Monitoring	End of EMU Stay
Oral and written Information	X			
Written consent		X		
Demographics		X		
Medical History		X		
Monitoring with the EpiSave system			Continuous	
Smartphone worn on the upper arm			Continuous	
VEEG			Continuous	
GCS Events			As occurred	
False Alarms			As occurred	
Skin check			Daily	
Smartwatch battery check & charging (by nurse)			Daily	
Technical Performance			Daily	
Data retrieval and erased from account			Daily	
Custom EpiSave smartphone App satisfaction survey completed by the nurse				X
Smartwatch wearability and comfort questionnaire (Patient-reported)				X
Smartphone wearability and comfort questionnaire (if applicable)				X
Device Return				X

Notes:

- All assessments are conducted during standard EMU care
- No additional visits or procedures beyond routine EMU care
- Device monitoring is continuous throughout EMU stay
- End-of-study assessments are conducted on the last day of EMU admission

Appendix 2: EpiSave System

Our EpiSave smartwatch app will be compatible with smartwatches running Wear OS, including the commercially available CE-marked TicWatch Pro 5 by Mobvoi used in this study.

The detection system will be based on 3D-acc measurements, with 32Hz sampling rate (can be upsampled from 25Hz or downsampled from 50Hz).

- **Application:** EpiSave GCS detection algorithm

- **Data Storage:** Data is initially stored locally on the smartwatch and automatically uploaded to a secure Swiss-based server – Exoscale – when the device is placed on its charger and connected to a Wi-Fi network with the Internet access or Bluetooth via the smartphone connection to the Internet.

- **Alert System:** The alert system is disabled for the participant during their stay in the EMU. When the EpiSave system detects a GCS, an alert is transmitted in real time to a dedicated smartphone/tablet held by the EMU nursing staff. Upon receiving the alert, the staff must verify whether a true GCS occurred by reviewing the VEEG data. If no GCS is confirmed—indicating a false positive – the staff should document the reason, such as the patient's activity or movement that may have triggered the alert. Conversely, if a GCS is observed in the VEEG but was not detected by EpiSave system (a false negative), the staff must report the missed event and describe the patient's activity and arm position before and during the GCS within the smartphone companion app.

- **Device Management:** The smartwatch used in this study is a commercially available Android device running Wear OS, such as the TicWatch Pro 5 from Mobvoi. Monitoring is continuous throughout the EMU stay. To avoid interruptions when the smartwatch needs to be charged, a second pre-configured device will be provided to the participant, ensuring seamless data collection.

Data recorded by the smartwatch is initially stored locally. Once the device is placed on its charger and connected either to a Wi-Fi network or Bluetooth, the data is automatically uploaded to the secure study server hosted by Exoscale in Switzerland.

At the end of the EMU stay, the smartwatch is retrieved, and any remaining data is downloaded to complete the data collection process.